datacat
Brightspeed Data, ML and AI workbench

DEEP DIVE · PURPOSE, PERSONAS, KEY FUNCTIONS AND THE REAL SCREENS · NO TECHNICAL BACKGROUND NEEDED

Everything our data, ML and AI teams need, on one governed workbench.

You open a browser, sign in with your normal Brightspeed account, and get the tools for your job already set up and already governed — a warehouse, pipelines, dashboards, a complete machine-learning lifecycle, and an AI assistant that can reach our own systems. Nothing to install.

Prefer paper? Download this deep dive as a PDF → · Full detail in the pages below.

Three ready-made seats: Data, ML and AI
One sign-in · one vault · one front door
Built from open source · no licence fees
Running today; Google Cloud is the target

THE SHORT VERSION

What it is, in three lines

Brightspeed's data, machine-learning and AI teams each get a ready-made “seat” — the same platform with the right tools already wired together. They share one sign-in, one encrypted place for credentials, one front door and one set of rules about what may leave the platform. It is built almost entirely from proven open-source software (dbt, Superset, MLflow, Jupyter, Ollama and more), so there are no licence fees and no lock-in.

SELF-SERVICE

Teams go from question to answer without waiting in a ticket queue.

GOVERNED

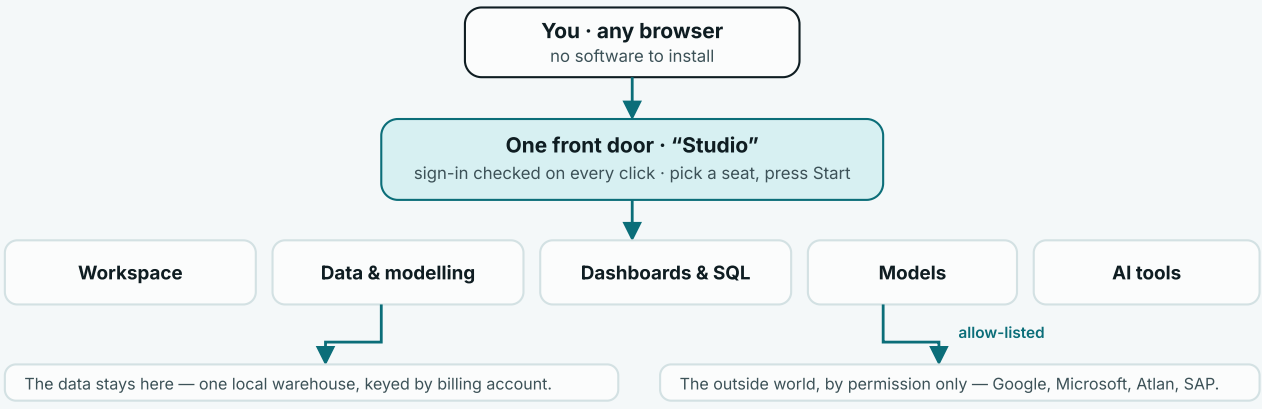
Every step is access-checked, logged and reproducible — speed without losing control.

READY FOR AI

An AI assistant that can query our real systems, under our rules and with a full audit trail.

HOW IT FITS TOGETHER

The whole thing in one picture



Read it as a house: one front door, a set of rooms for each job, your records kept inside, and a single guarded exit to the outside world.

WHO IT'S FOR

Three seats, one platform

A “seat” is a preset of the same platform. All three run side by side, sharing the same controls and rules.

DATA PROFESSIONAL

Land it, model it, show it

For data engineers, analytics engineers and data stewards.

- No-code pipelines, including on-premise databases
- dbt models and automatic data tests
- Superset dashboards, Power BI, cross-source SQL
- An Atlan catalogue connector for findability

ML SCIENTIST

From notebook to production

For data scientists.

- Explore, choose features, test hold-out strategies
- Track experiments and compare model families
- Release checks, a second pair of eyes, rollback
- Drift monitoring, retraining, recorded evidence

AI ENGINEER

Put AI to work on real tasks

For AI and automation engineers.

- A gateway over cloud and local models
- Tools that reach our real systems, under our rules
- Disputes → SAP, and call tagging, out of the box
- Runbook search and a familiar chat app on top

KEY FUNCTIONS AT A GLANCE

What goes in, what comes out

Each row is a capability you can use on day one. The last column names the artifact to look for on screen; the tour below shows the live screens.

FUNCTION	WHY IT MATTERS	HOW IT WORKS	INPUT → OUTPUT / RESULT
Integration	Reach on-premise and SaaS data without writing a pipeline.	Integration canvas (dlt), incremental runs, scheduling.	Files, Postgres/Oracle/SQL Server, Microsoft 365, REST, BigQuery, Atlan → governed raw tables, run history and lineage.
Model & test	One trusted definition of every metric.	dbt Studio: staging → intermediate → marts, tests and docs.	Raw tables → tested marts, documentation, data-quality results and column lineage.
Analytics & SQL	Answer questions without copying data into another tool.	Superset for published dashboards; Trino for one SQL dialect across sources.	Marts and federated sources → certified dashboards, ad-hoc SQL, one number everyone can trace.
ML lifecycle	Promote models on evidence, not opinion.	MLOps Control Tower + MLflow: features, experiments, gates, serving, drift, retrain.	Features and labels (e.g. 90-day churn) → champion/challenger, gate results, model card, drift report, retrain record.
Agentic AI	Let AI act on real systems under our rules.	AI Harness + MCP tools; browser-backed BigQuery/Atlan/Power BI fallbacks.	Question + approved tool groups → grounded answer, tool calls, audit trail and, where needed, a visible browser action.
Knowledge / RAG	Use what we already wrote instead of guessing.	RAG collections + Open WebUI.	Runbooks, designs and policies → cited answers, collection and chunk statistics.
Catalogue & lineage	Find the right asset and know what depends on it.	DataHub catalogue (optional) + Atlan connector.	Tables, models, dashboards and jobs → search results, owners, tags and upstream/downstream lineage.
Disputes & calls	Automate high-volume work with a human checkpoint.	Dispute Desk; CCAI call ingestion and tagging.	Mailbox or call logs → structured case with confidence and SAP filing, or tagged call reasons, sentiment and churn signals.
Governance	Self-service without losing control.	Studio SSO, encrypted vault, allowlisted egress, per-component cloud identity, audit log.	Identity, secrets and policy → signed-in access, injected credentials, blocked egress and a person-level audit trail.

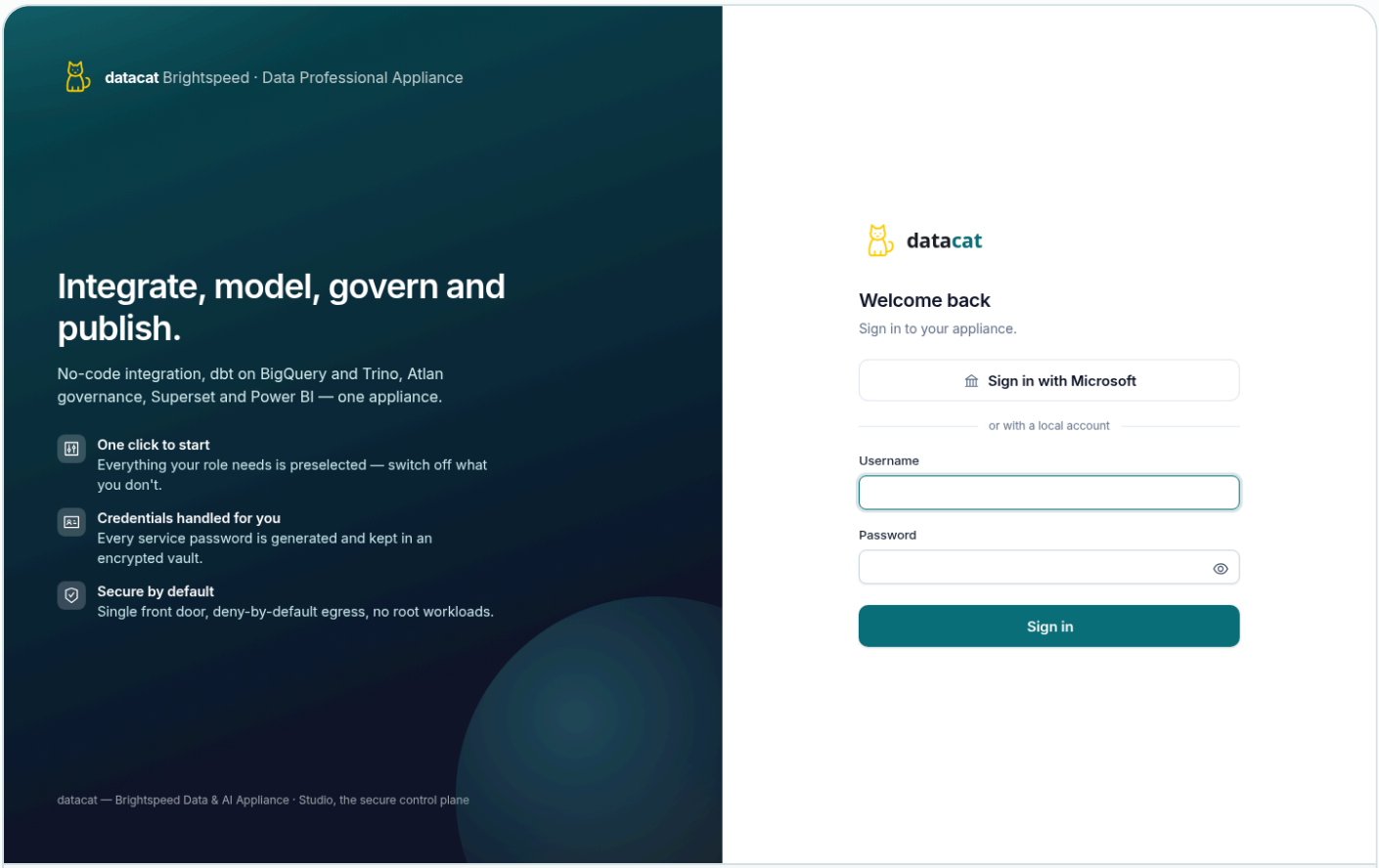
WHAT YOU'LL SEE

A tour of the real screens

Every screen below is from the running workbench — nothing here is a mock-up.

01 Sign in

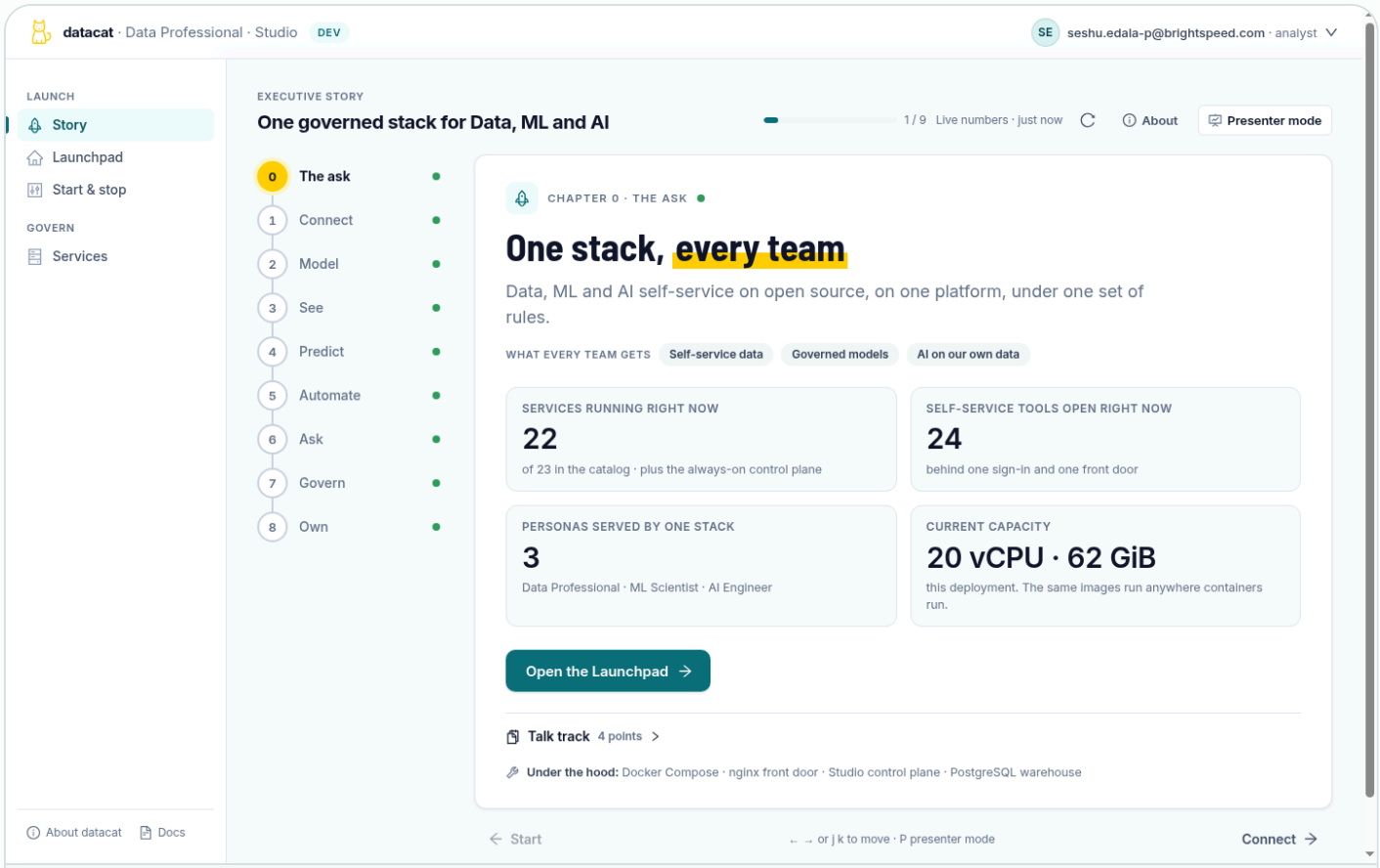
Your usual Brightspeed (Microsoft) account gets you in. Administrators are named on a list, not handed a shared password.



Sign in. One front door for every tool — the same page whatever your role.

02 The guided tour

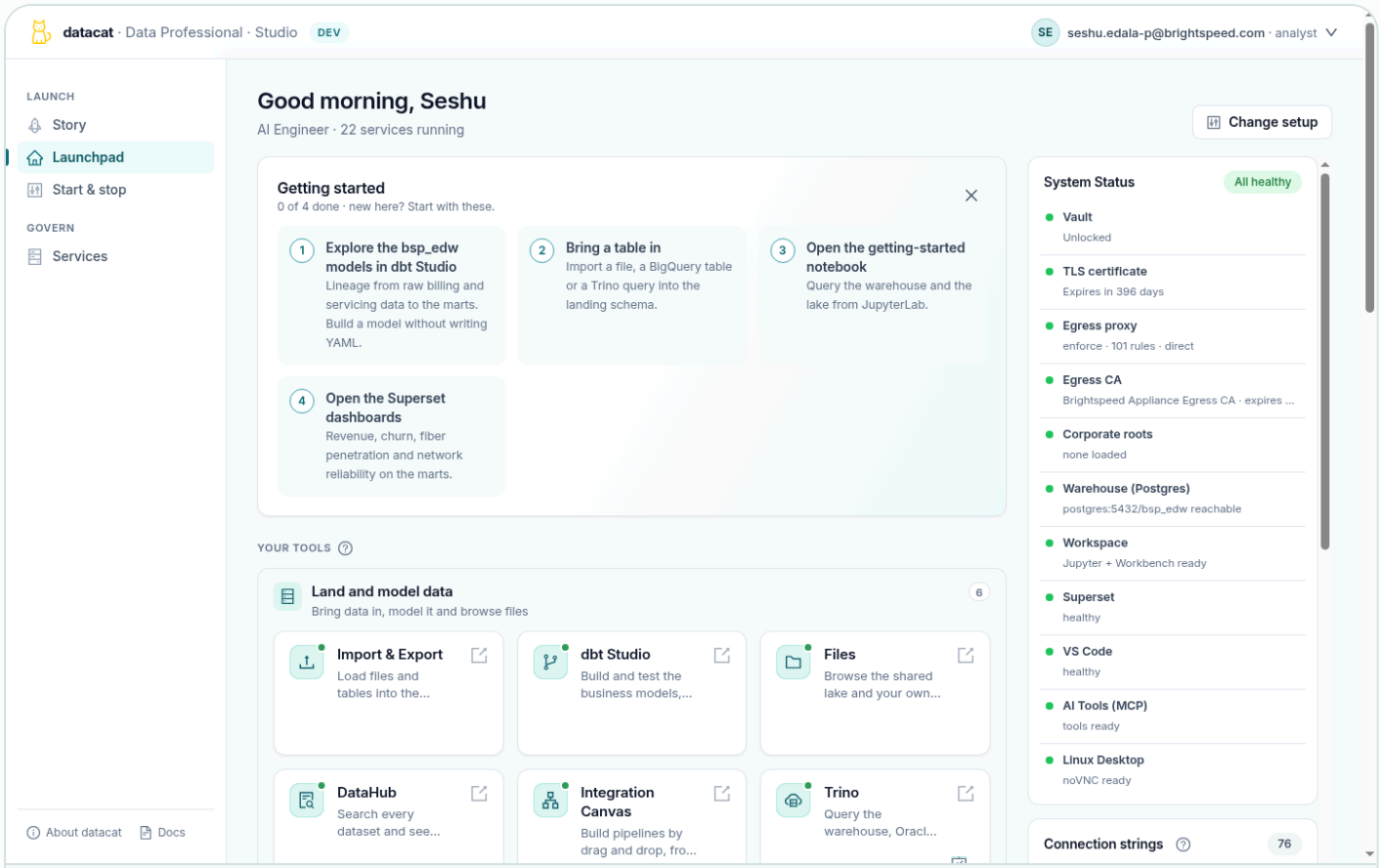
Studio opens on **Story**, the workbench's own walkthrough. It reads the live platform — services running, tools, seats — so the numbers on screen are the ones in your appliance, not a slide.



Story. A guided tour that uses live numbers from the running appliance.

03 Your launchpad

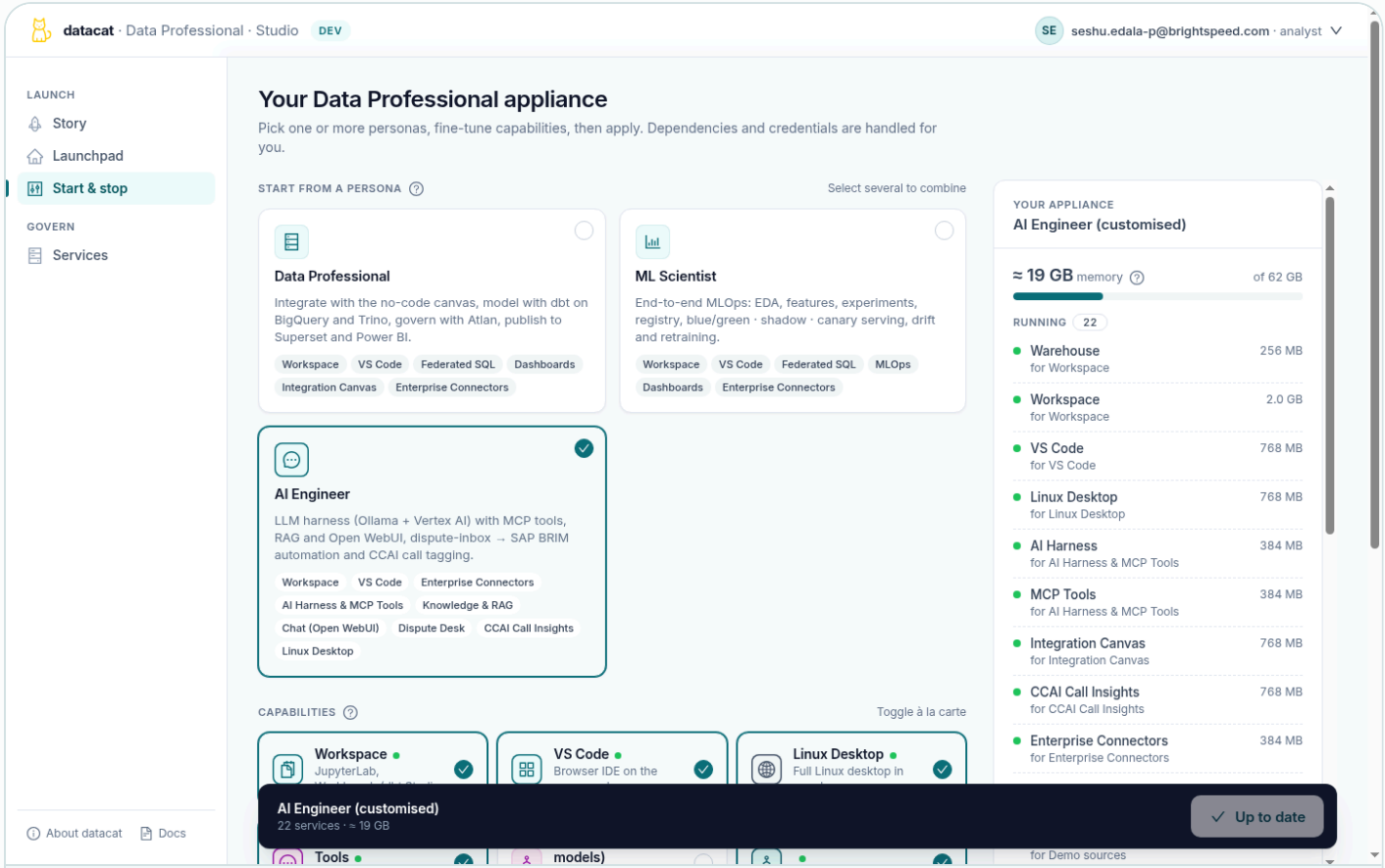
After signing in you land on Studio's Launchpad: a card for every tool, grouped for your job, with the health of the platform on the right and connection details you can copy — you never need to ask anyone for a password.



Launchpad. Everything that is running, grouped by role, with live system status.

04 Pick your tools

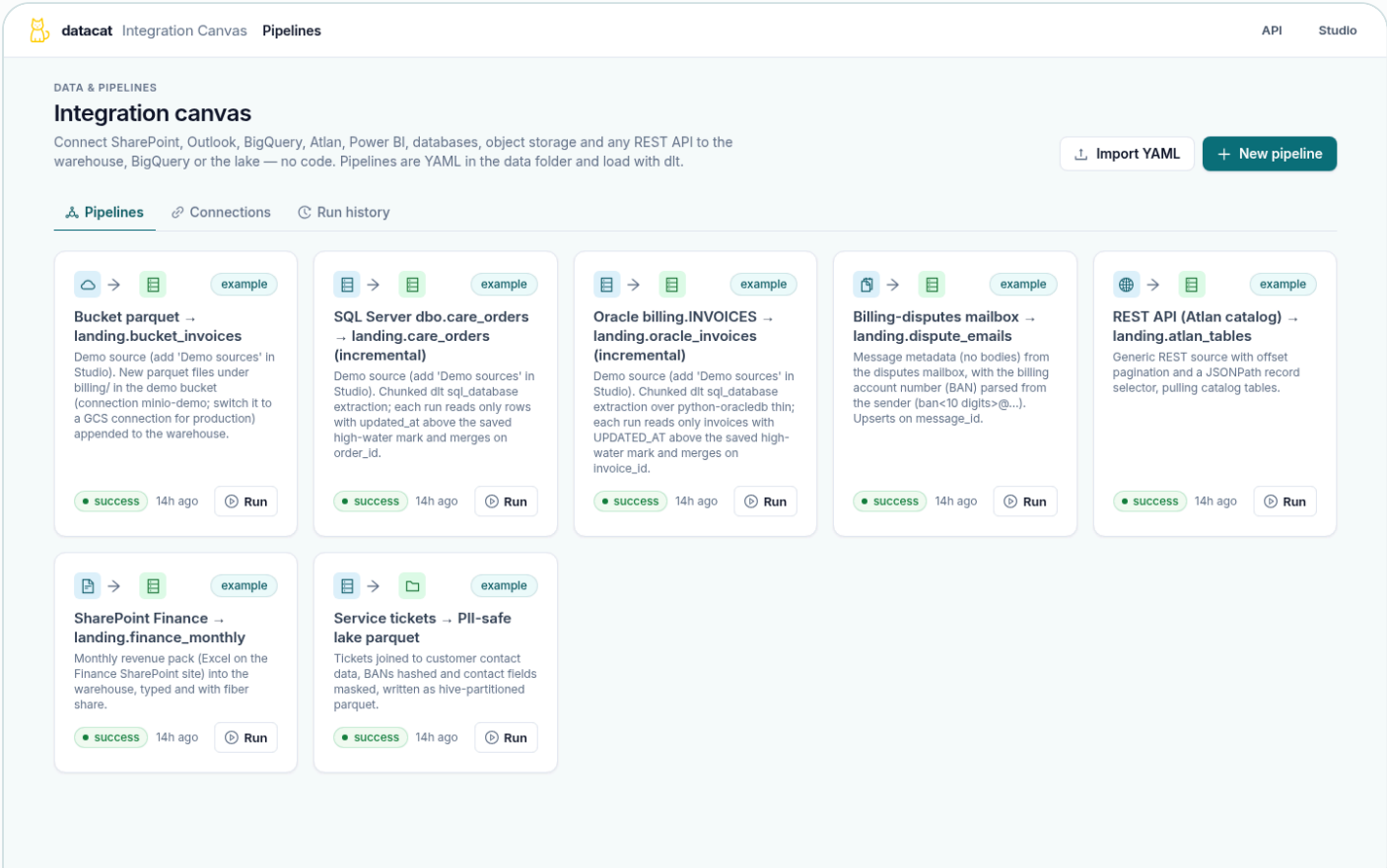
Start from a preset — Data Professional, ML Scientist or AI Engineer — or fine-tune the capabilities. The platform adds what they depend on and starts everything for you.



Start & stop. Choose a seat or fine-tune it, then apply. The right-hand panel shows memory and every running service.

05 Bring data in — without code

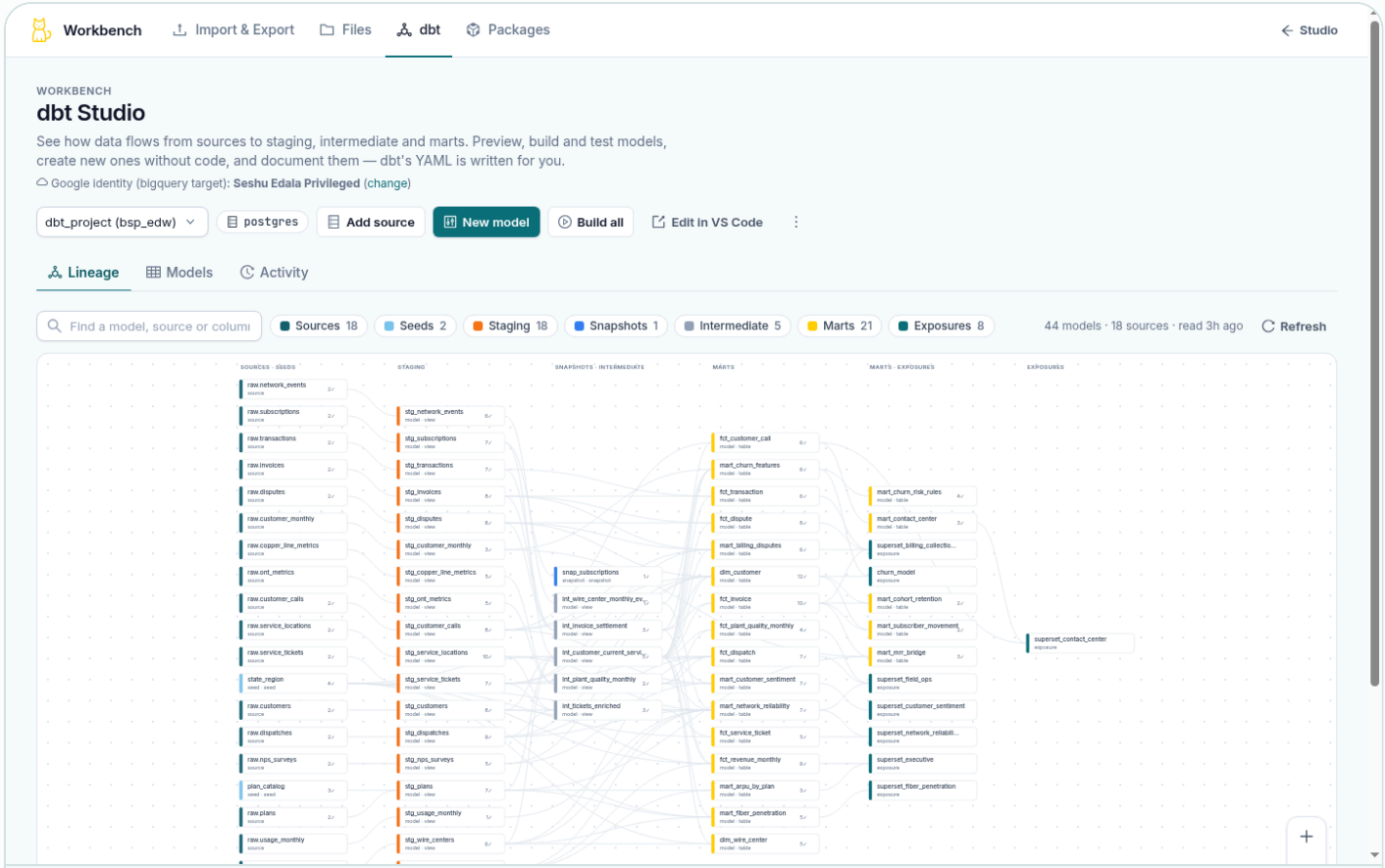
The integration canvas connects files, databases (including on-premise ones), Microsoft 365, REST APIs, BigQuery and Atlan to the warehouse. Each pipeline shows its last run and loads only what has changed.



Integration canvas. No-code pipelines from our systems into the warehouse — each with a run button and a last-run status.

06 Model the data, and document it

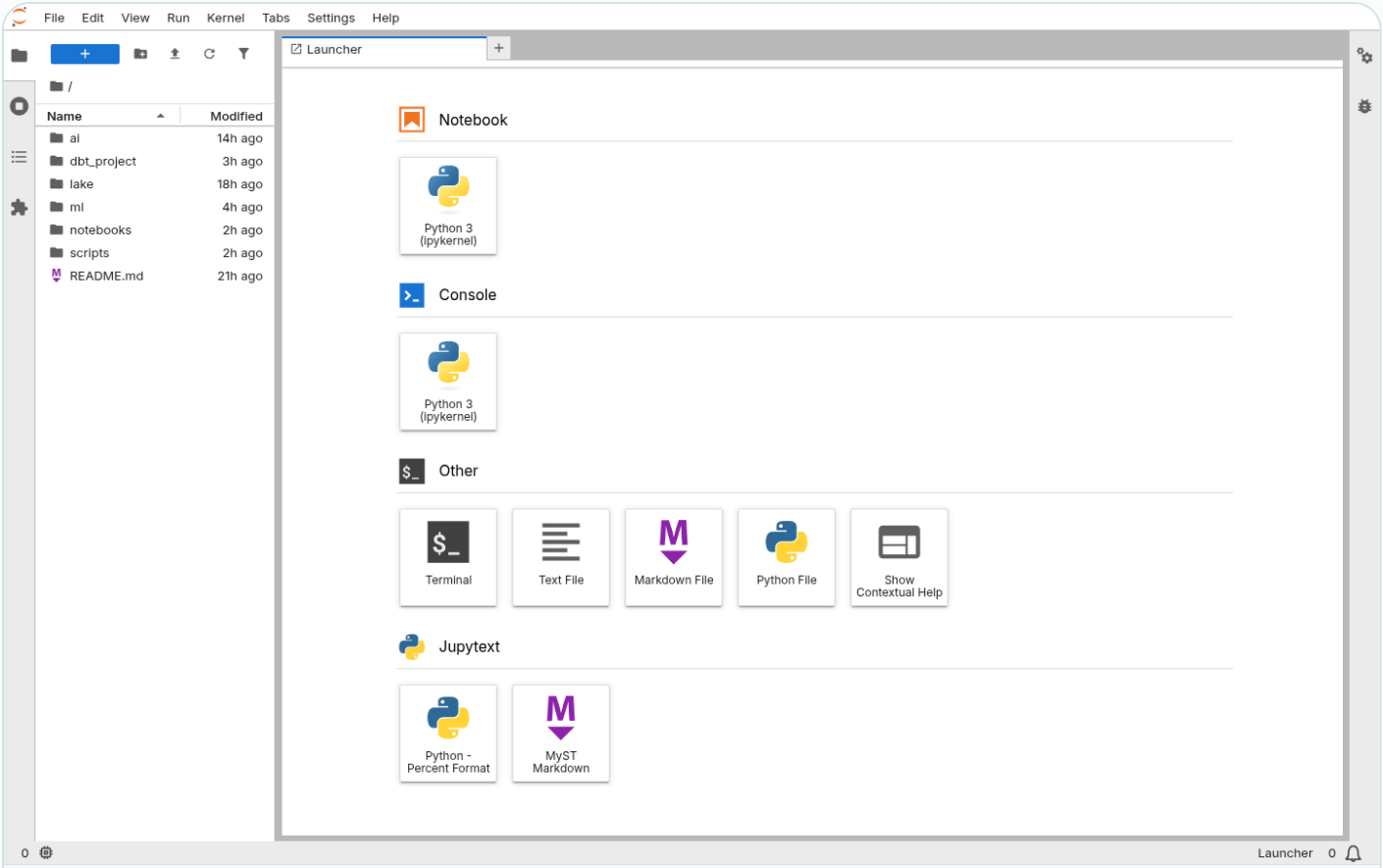
dbt turns raw data into clean, tested tables and draws the lineage from source to finished metric. Anyone can follow where a number came from.



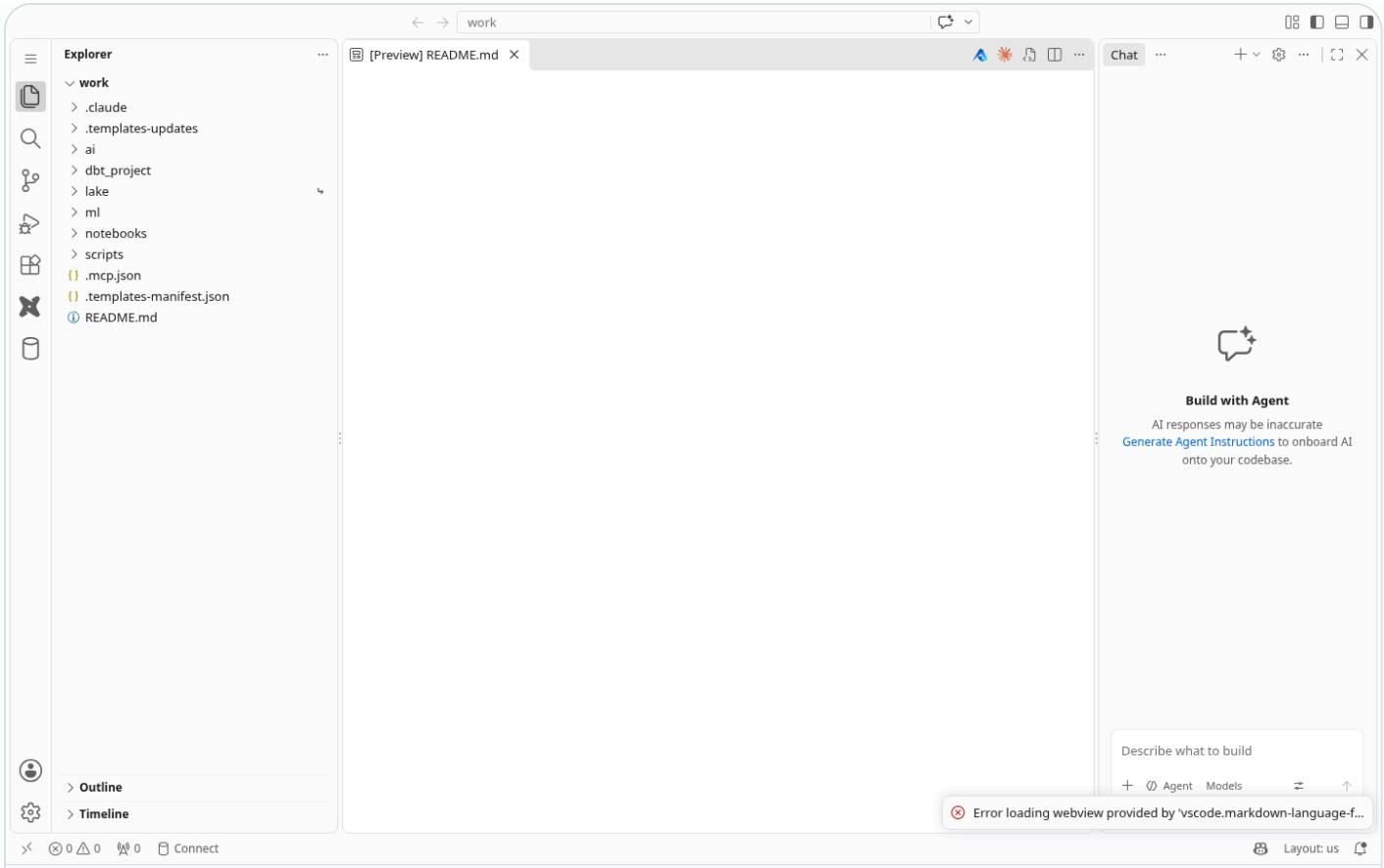
dbt Studio. From raw sources to trusted tables — every step visible and documented.

07 Your workspace

Two ways into the same files: JupyterLab for notebooks and a full VS Code in the browser for projects and scripts. Both are already wired to the warehouse and to your workspace folder.



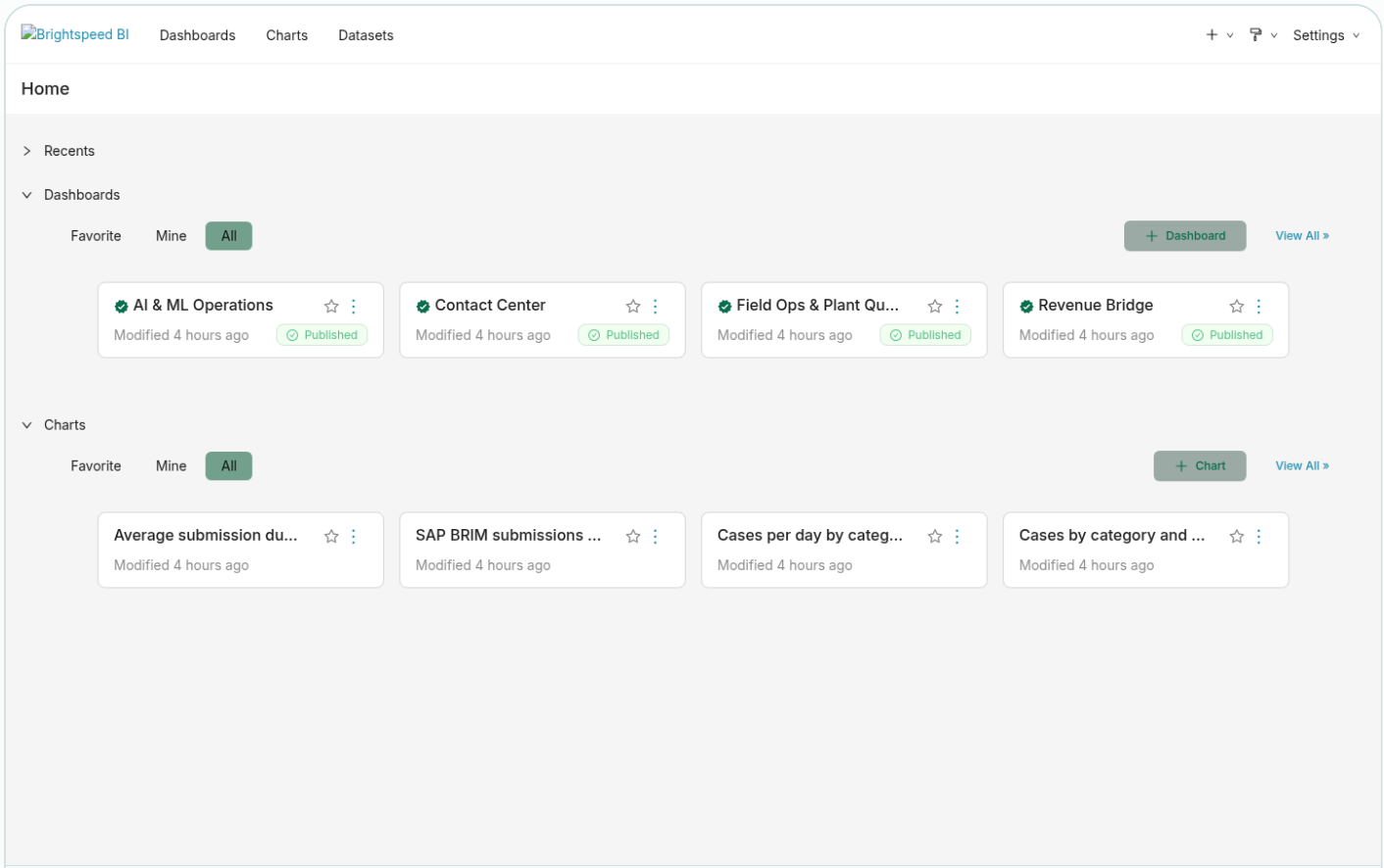
JupyterLab. Notebooks, consoles and terminals against the governed warehouse.



VS Code. The same files in a full editor, with an AI assistant beside them.

08 Dashboards and SQL

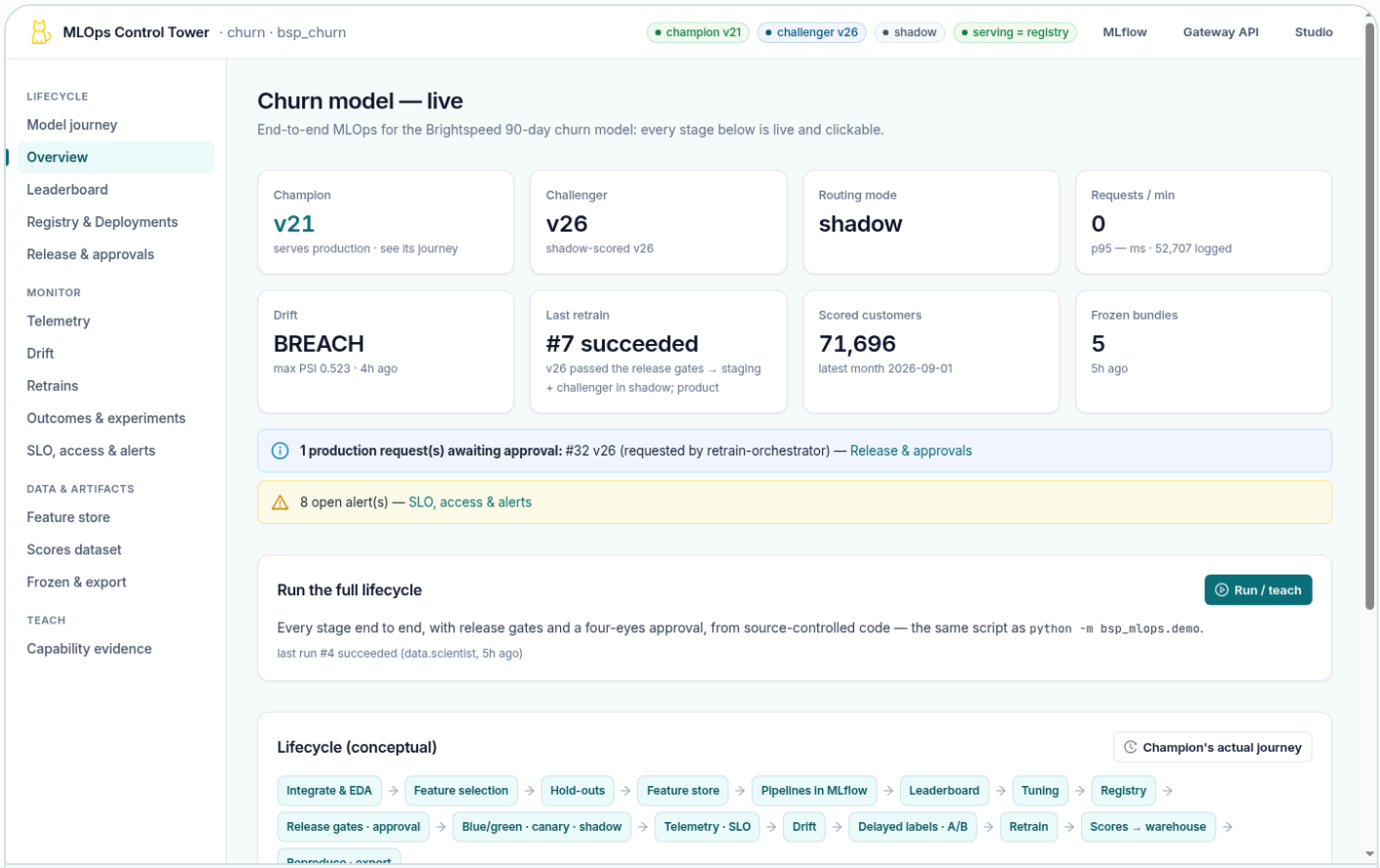
Superset publishes the dashboards — revenue, field operations and plant quality, contact centre, and the AI & ML operations board — over the same governed warehouse. Trino gives any tool one SQL dialect across sources.



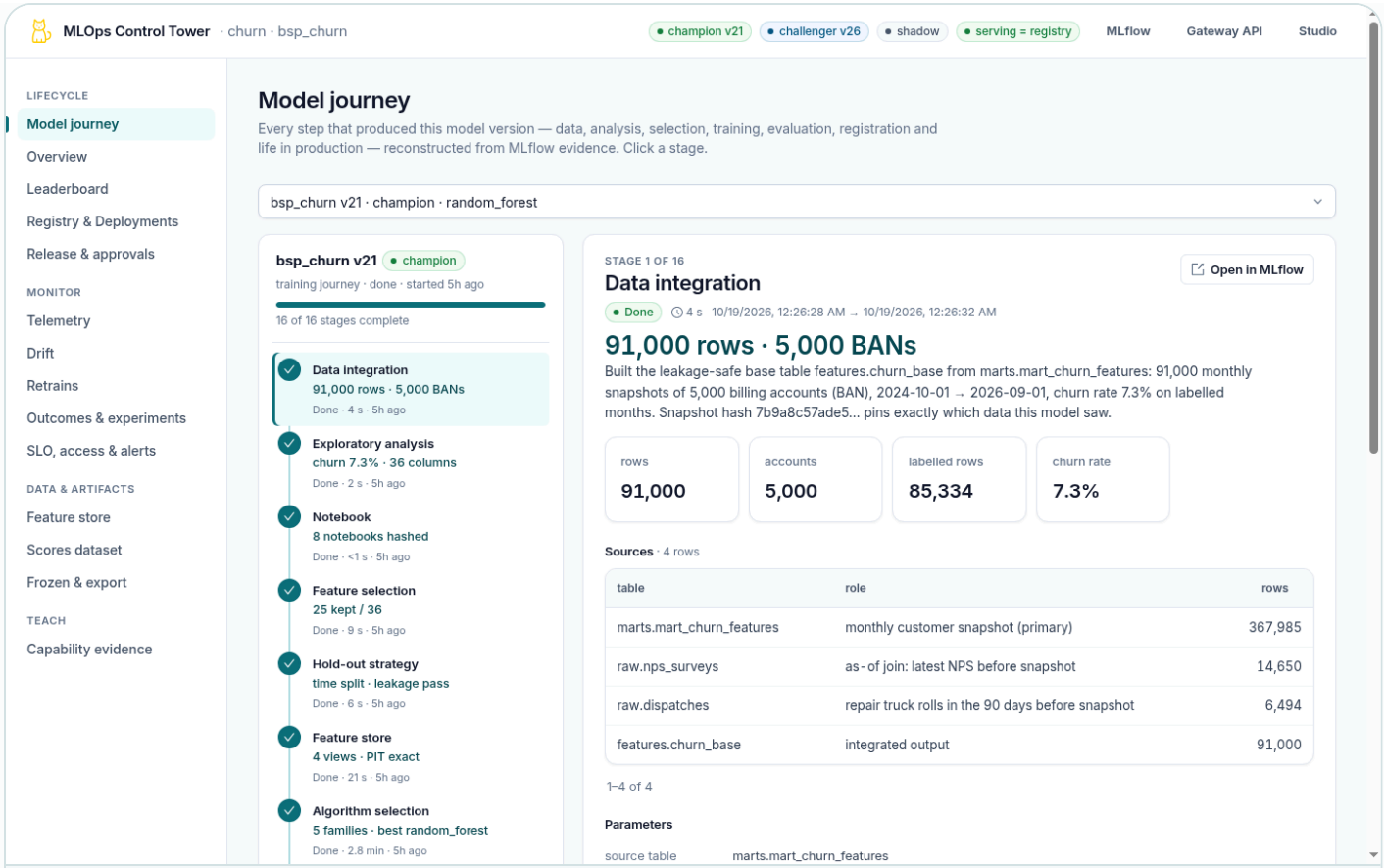
Superset. Published dashboards for each business area, over the same governed data.

09 Take a model from notebook to production

The MLOps control tower runs one real use case — 90-day customer churn. It tracks every experiment, promotes a model only after automatic checks (and a second person) pass, serves it safely, watches for drift, retrains, and can roll back.



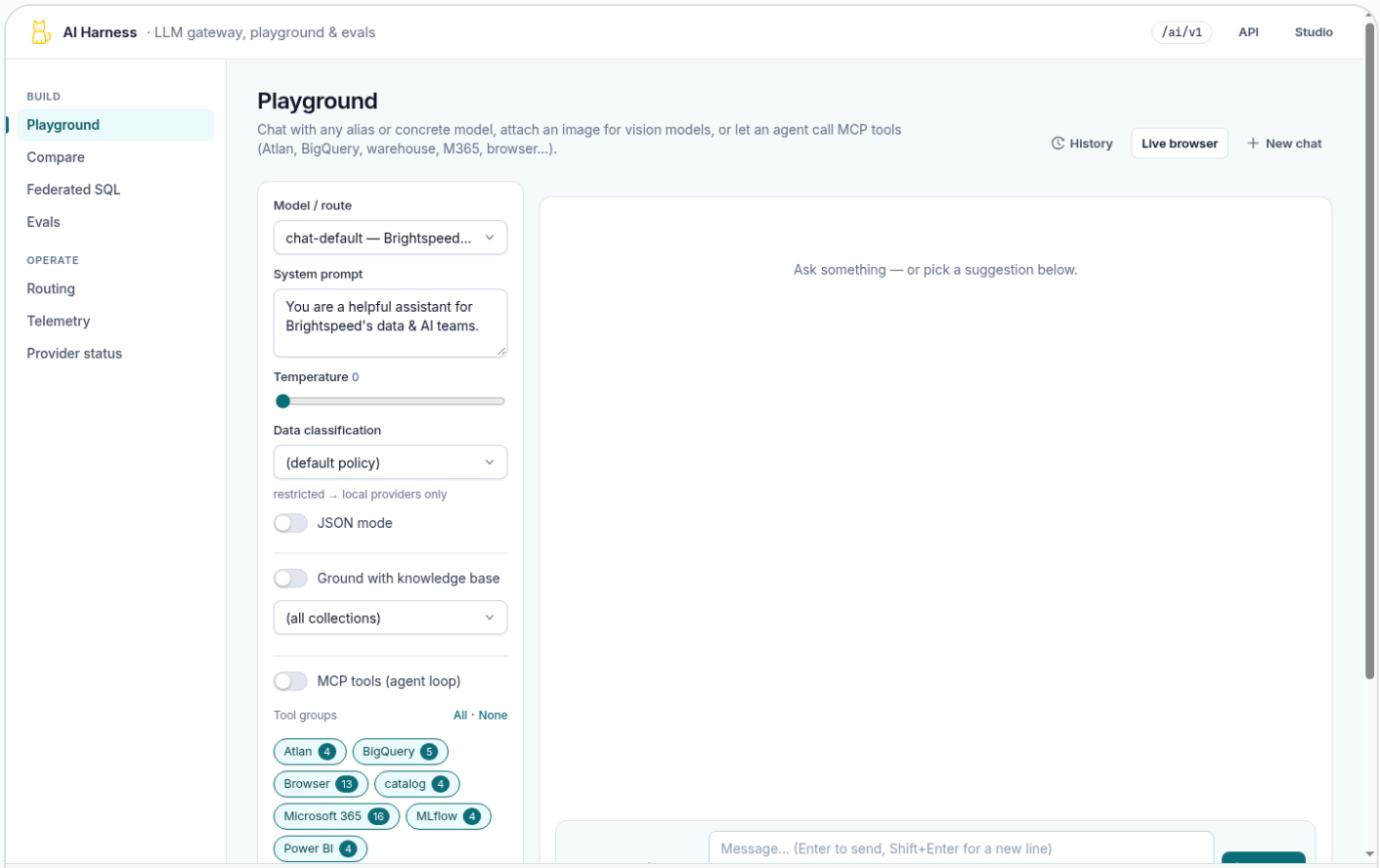
MLOps control tower. Champion, challenger, serving mode, drift and retraining on one page.



The model's journey. Every step that produced the version in production, with the evidence — nothing is a slide.

10 Put AI to work

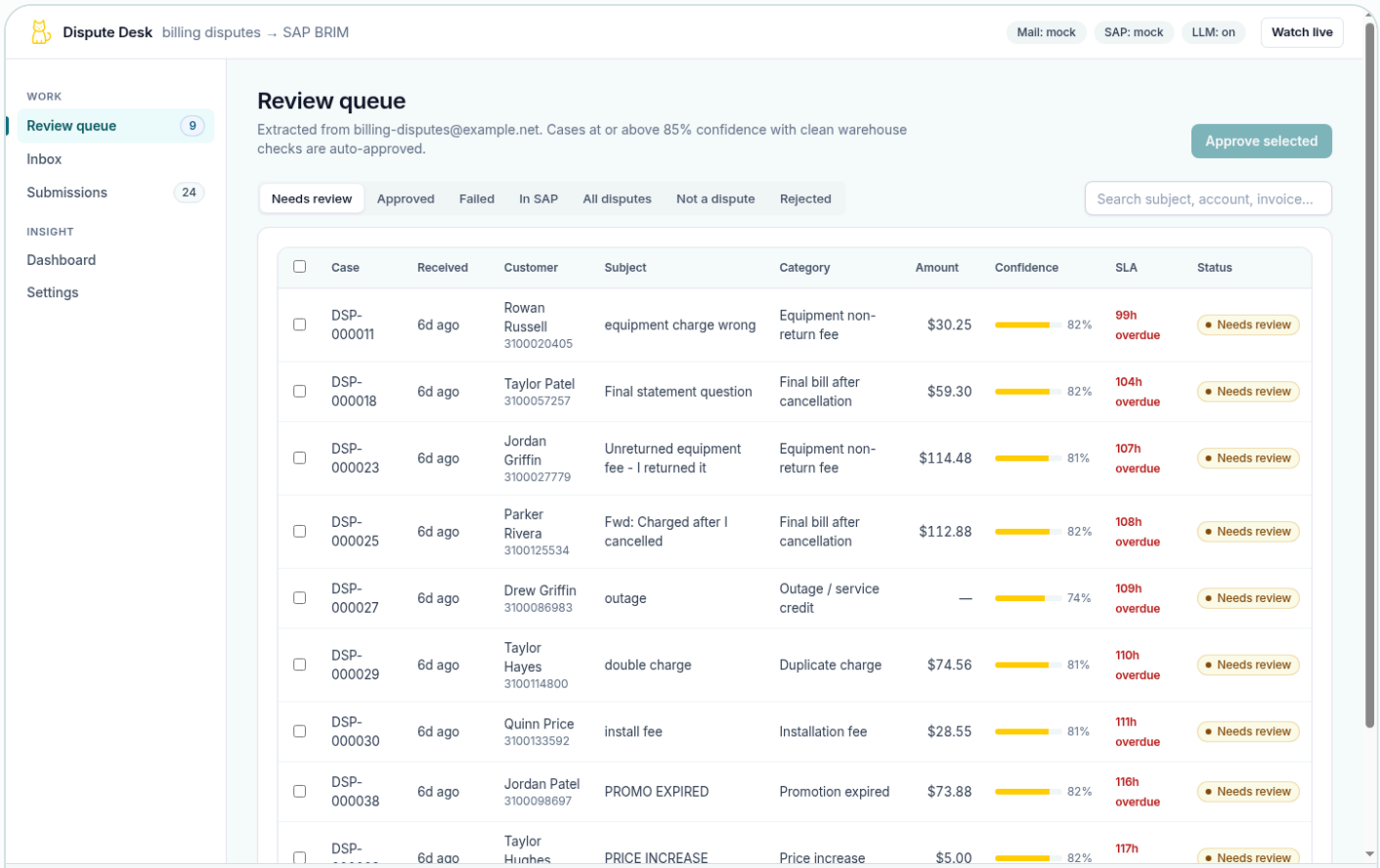
The AI harness is a playground over cloud models (Google Gemini) or a local model. Switch on tools and the assistant can query the warehouse, BigQuery, Atlan, Power BI and Microsoft 365 — and you can watch a browser doing the work.



AI playground. Choose a model, set the rules, switch on the tools it may use.

11 Automate a real task: billing disputes

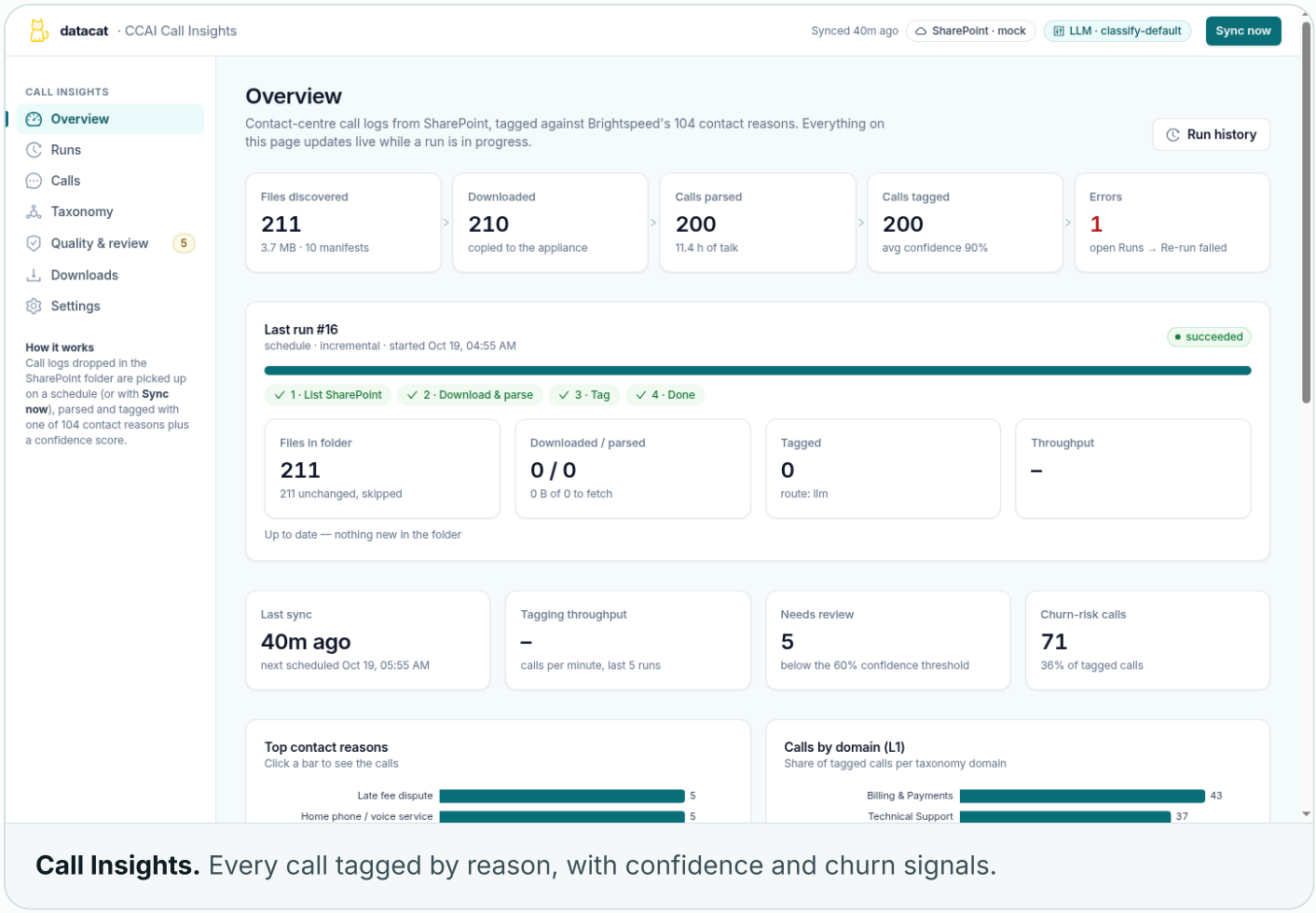
Customer dispute e-mails are read, turned into structured cases, checked against our records, and filed in SAP — automatically when confident, or after a person approves. A reviewer can watch the browser fill in the SAP form step by step.



Dispute Desk. Each e-mail becomes a validated case with a confidence score, ready for SAP.

12 Turn calls into insight

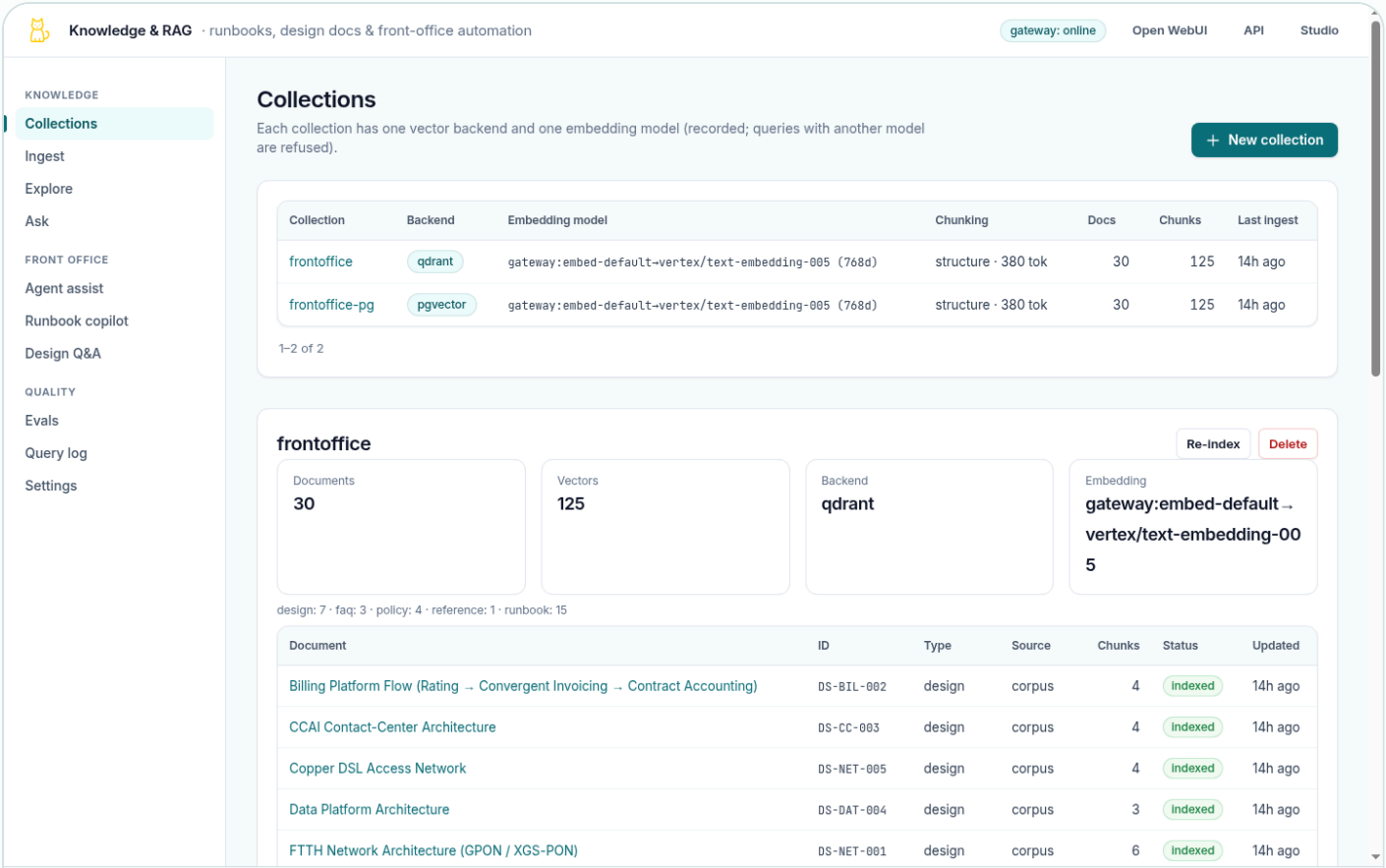
Contact-centre call logs are downloaded, parsed, and tagged against Brightspeed's contact reasons with a confidence score — giving managers a live picture of why customers call, and which calls signal churn.



Call Insights. Every call tagged by reason, with confidence and churn signals.

13 Ask our own documents

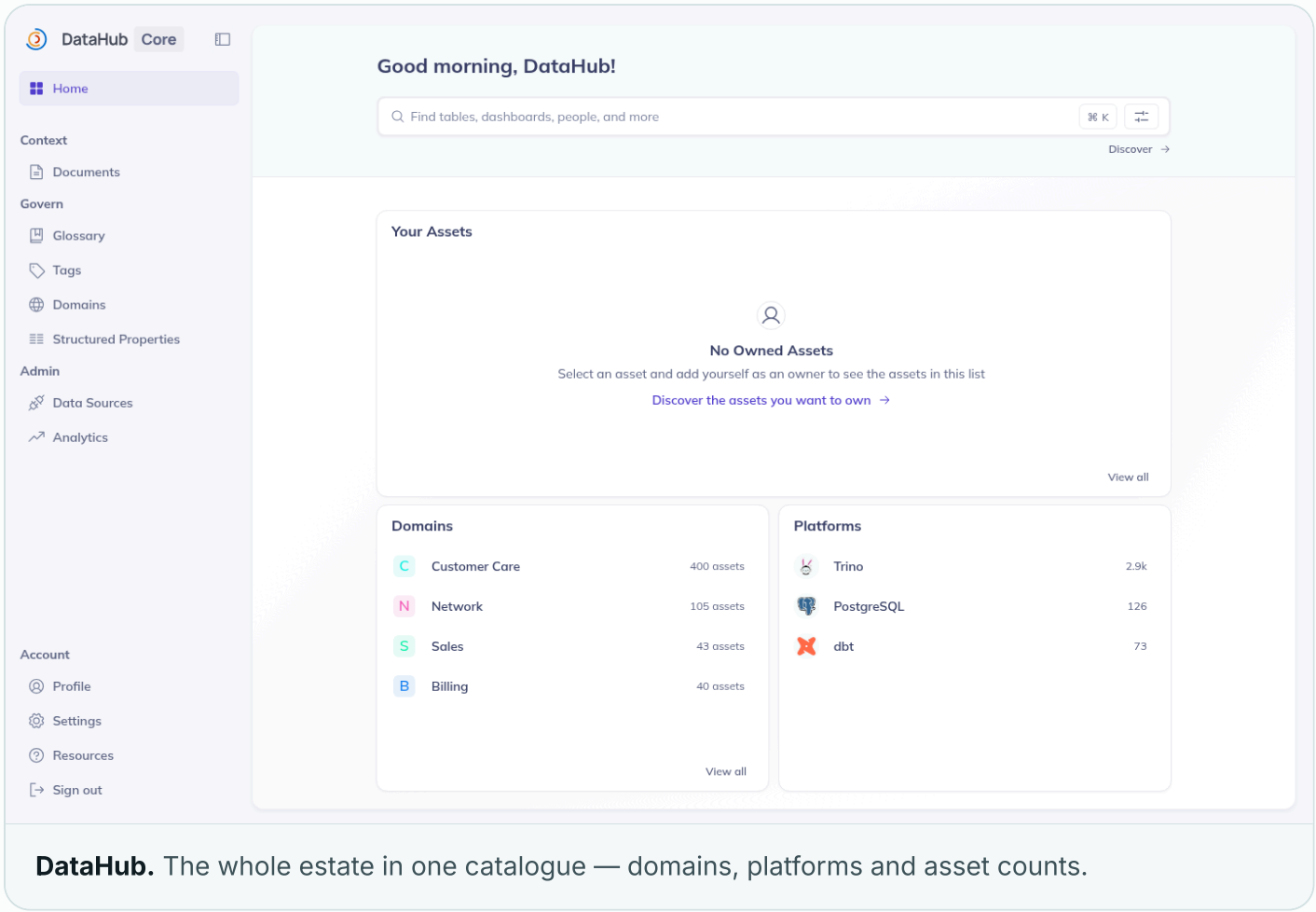
The knowledge search (RAG) indexes our runbooks and design docs, so questions are answered from what we actually wrote — with citations, not guesses.



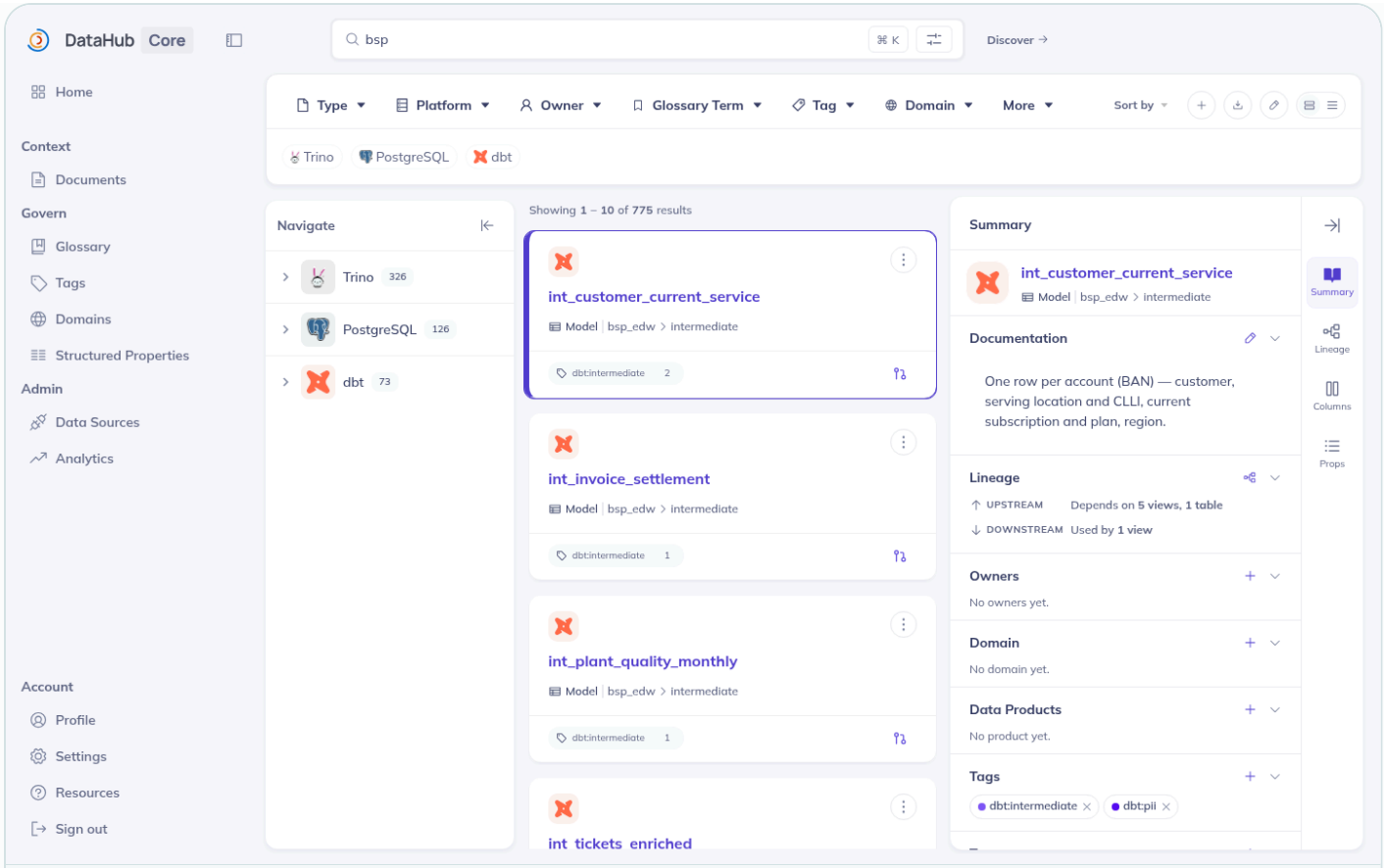
Knowledge & RAG. Our runbooks and designs, searchable, with grounded answers.

14 Catalogue and lineage

DataHub is the catalogue layer: every dbt model, table and dashboard, grouped by domain — Customer Care, Network, Sales, Billing — with owners, tags, documentation and upstream/downstream lineage. Search finds an asset; the lineage shows what depends on it.



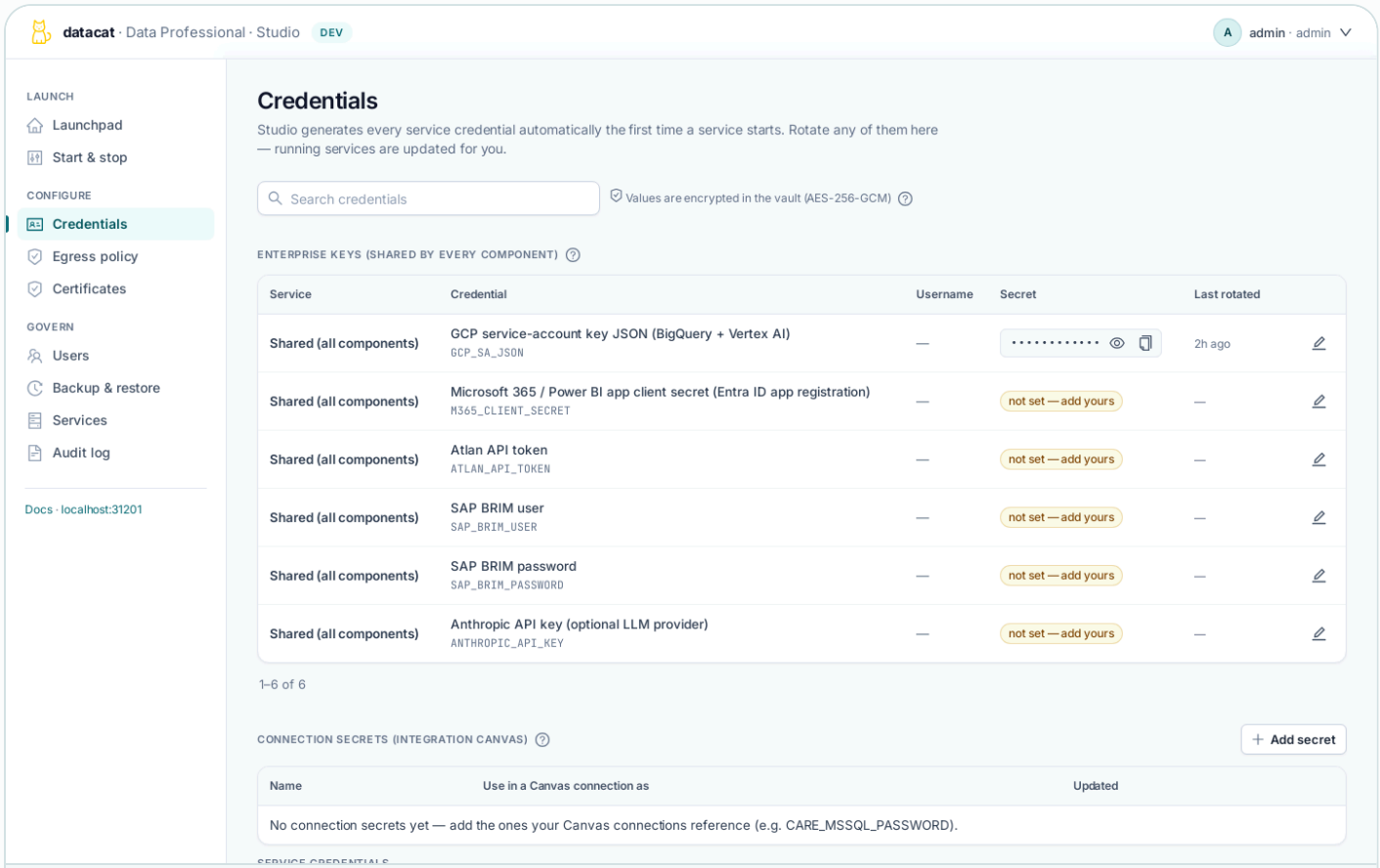
DataHub. The whole estate in one catalogue — domains, platforms and asset counts.



Search and lineage. Find a model, then see what it depends on and what uses it.

15 Keys and rules, in one place

Every credential lives in one encrypted vault. Tools receive them automatically; they are never written into code or files. The same console controls who may sign in and what the platform is allowed to reach.



Credentials & vault. One encrypted safe for every key and password.

SELF-SERVICE, STILL CONTROLLED

The guardrails

Brightspeed sign-in
Microsoft single sign-on for every tool; admin rights come from a named list.

Encrypted credentials
Every service gets its own random password, kept in one encrypted vault.

Outbound allowlist
Traffic can leave only for approved hosts; file-sharing and personal mail are blocked on purpose.

Per-component cloud identity
Each capability can use a different Google identity and project.

Audit trail
Sign-ins, credential changes, approvals and AI tool calls are logged with the person behind them.

Restricted data stays local
The AI gateway sends restricted prompts only to the local model, and logs every call.

GETTING IN

How to start

You need three things: your Brightspeed account, the workbench address your administrator sends you, and a modern browser on the corporate network or VPN.

1. **Open the address and sign in**
   Use **Sign in with Microsoft** and your usual account. The first page is Studio; everything opens from there. Bookmark it.

2. **Find your way around**
   The **Launchpad** shows a card for every running tool. **Docs** is the full manual. **Story** is the guided tour — the best place to start.

3. **Do your first hour**
   Open the starter notebooks for your seat. Everything uses the synthetic demo data, so nothing you do can break anything that matters.

4. **Keep your work**
   Your files live in your workspace folder and survive restarts. Keep credentials in the vault, not in notebooks, and push your projects to git.

> **Don't have access yet?** Ask your appliance administrator (or the platform team) to add you to the workbench group. New here? Read the full manual at /docs/.

This deep dive as a PDF → | Full manual →

Screens are from the running workbench. The demo dataset uses generated copper and fiber account data, not customer records.